



**Audencia Business
School**

**2026
March**

Marketing Analytics Project

Advanced Analytics for Customer
Segmentation and Sales Forecasting

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1. Executive Summary

This project analyses the retail performance and customer behaviour of **two distinct datasets**:

The **US Superstore** (2015–2018) and **Adidas US Sales** (2020–2021), to develop a holistic view of modern retail dynamics. Working across 793 unique Superstore customers and approximately 9,648 daily Adidas sales records, our group applied a five-stage analytical pipeline designed to move from customer understanding to operational action.

The methodology follows a deliberate logical chain:

- RFM Scoring -> K-Means Segmentation -> LDA Validation -> CLV Prediction -> Sales Forecasting.
- Each stage builds on the previous, ensuring that insights are not isolated findings but connected inputs into a unified decision-making framework.

Key findings reveal significant commercial concentration: just **8%** of Superstore customers generate **37.5%** of total profit, and the **top 20%** of customers account for **65.2%** of historical positive profit. **LDA** validated the customer segments with **95.8%** accuracy, confirming that **Discount** behaviour and **Purchase** Frequency are the **two primary drivers** of segment differentiation. The **CLV** model further demonstrated that Priority **A customers** (18.9% of the base) capture **27.4%** of predicted future value, supporting a **value-driven** rather than volume-driven marketing strategy.

On the **forecasting** side, the **Holt-Winters** model successfully captured seasonal demand cycles for **new product planning**, while XGBoost outperformed LSTM for Adidas daily sales prediction: a finding explained by the relatively small dataset size, which limits deep learning effectiveness.

Our **recommendations** centre on **four** priorities: **protecting** and **retaining high-CLV** customer segments through non-discount tools, reforming discount allocation using **RFM** tier rules, redirecting promotional **investment** toward Technology categories, and building procurement buffers aligned to seasonal demand peaks. Together, these actions provide a framework for **more efficient inventory management**, targeted campaigns, and measurable profit growth.

2. Marketing Decision Objectives

To move the needle on profitability, this analysis is structured around **four core decision pillars**, each mapped to a specific analytical method:

- **Identify High-Value Segments:** Categorise customers based on their actual purchasing behaviour — recency, frequency, spend, and discount usage — to determine where marketing budget will generate the highest return. This is addressed through RFM Feature Construction and K-Means Clustering, validated by Linear Discriminant Analysis.
- **Predict Future Revenue:** Move beyond historical reporting to forward-looking profit prediction. By applying Customer Lifetime Value modelling with rolling temporal snapshots, the analysis enables prioritisation of customers based on their expected future contribution rather than their past spend alone. For new product categories, Holt-Winters exponential smoothing provides a seasonal demand benchmark to guide launch timing.
- **Optimize Product Strategy:** Analyse category-level sales to identify which product lines destroy margin (notably Furniture) and which should receive increased promotional investment (notably Technology). This directly informs cross-sell targeting for the highest-value customer segments.
- **Reduce Revenue Leakage:** Evaluate the structural impact of product returns on net profit. By explicitly removing returned orders before calculating customer-level profit metrics, the analysis ensures that all CLV scores, segment profit averages, and forecasts reflect true retained revenue — not inflated gross figures that would overstate business health.

These **four pillars are not independent** — they form an integrated decision chain. Segment identification informs CLV prioritisation; CLV prioritisation shapes recommendation strategy; sales forecasting operationalises the timing of those recommendations across the planning calendar.

3. Data Sourcing and Quality

The analysis is built upon **three primary datasets**:

- The **Superstore Orders Data** provides a detailed view of US-based retail operations spanning January 2015 to December 2018, covering 9,994 order rows across 793 unique customers, 3 product categories (Furniture, Office Supplies, Technology), and 4 geographic regions (West, East, Central, South).
- The **Superstore Returns Data** contains 296 returned orders, which were merged with the Orders table to calculate **Net Sales** — ensuring profit projections reflect actual retained revenue rather than inflated gross figures.
- The **Adidas US Sales Dataset** contributes specialized insights into athletic footwear and apparel sales across 6 retail channels (e.g., in-store, outlet, online) and 5 product categories, covering the period January 2020 to December 2021 with approximately 9,648 rows of daily sales records.

Data quality was ensured through rigorous preprocessing. We addressed missing values, corrected date formats, and merged the Orders and Returns tables to create a unified Superstore view. Returned orders were explicitly flagged and excluded from profit calculations to avoid overstating performance. For the Adidas dataset, outlier values were removed and the date column was standardised to a consistent time format to support time-series feature engineering downstream.

4. Methodology & Analysis

Five methods were selected because they form a logical chain from customer understanding to operational action. The diagram below shows how each method feeds into the next.

- Method chain: RFM Scoring → K-Means Segmentation → LDA Validation → CLV Prediction → Sales Forecasting
- RFM quantifies individual behaviour. K-Means groups it. LDA validates the groups. CLV values them over time. Forecasting operationalises the timing.

4.1. Segmentation (RFM Feature Construction & K-Means Clustering)

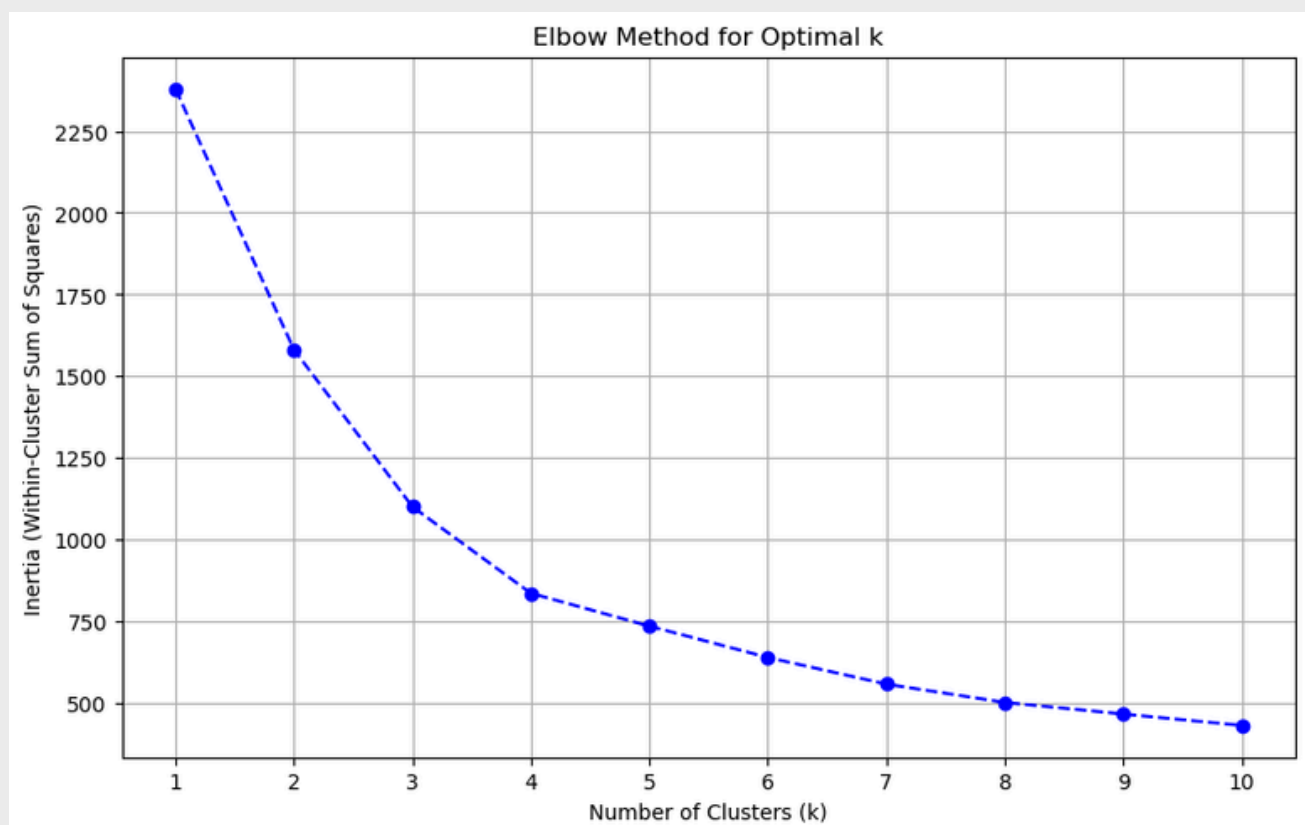
Objective: The marketing manager needs to allocate budget across customer groups. Without segmentation, every customer looks the same. K-Means is well-suited here because it groups customers by revealed behaviour — what they actually bought, when, and how much — rather than demographic proxies. This produces segments that can be acted on immediately with existing CRM data.

How we applied it

We built an RFM profile for each of the 793 unique customers in the Superstore dataset: Recency (days since last purchase), Frequency (number of distinct orders), and Monetary value (total spend). We also included Total Profit to distinguish high-revenue from high-profit customers, and average Discount rate to capture price sensitivity. Features were standardised before clustering so that dollar values did not dominate over order counts. The optimal number of clusters was determined using the Elbow Method.

RFM Analysis assigns every customer a scored rank on three independent axes — how recently they bought, how often they buy, and how much they spend — producing an interpretable per-customer score that can be used directly for campaign selection, without requiring cluster membership.

RFM also serves as the foundation for all downstream methods in this project. Without computing these three metrics at customer level, neither segmentation nor CLV estimation is possible. It is therefore both a standalone analytical output and a prerequisite for the rest of the pipeline.



4.1. Segmentation (RFM Feature Construction & K-Means Clustering)

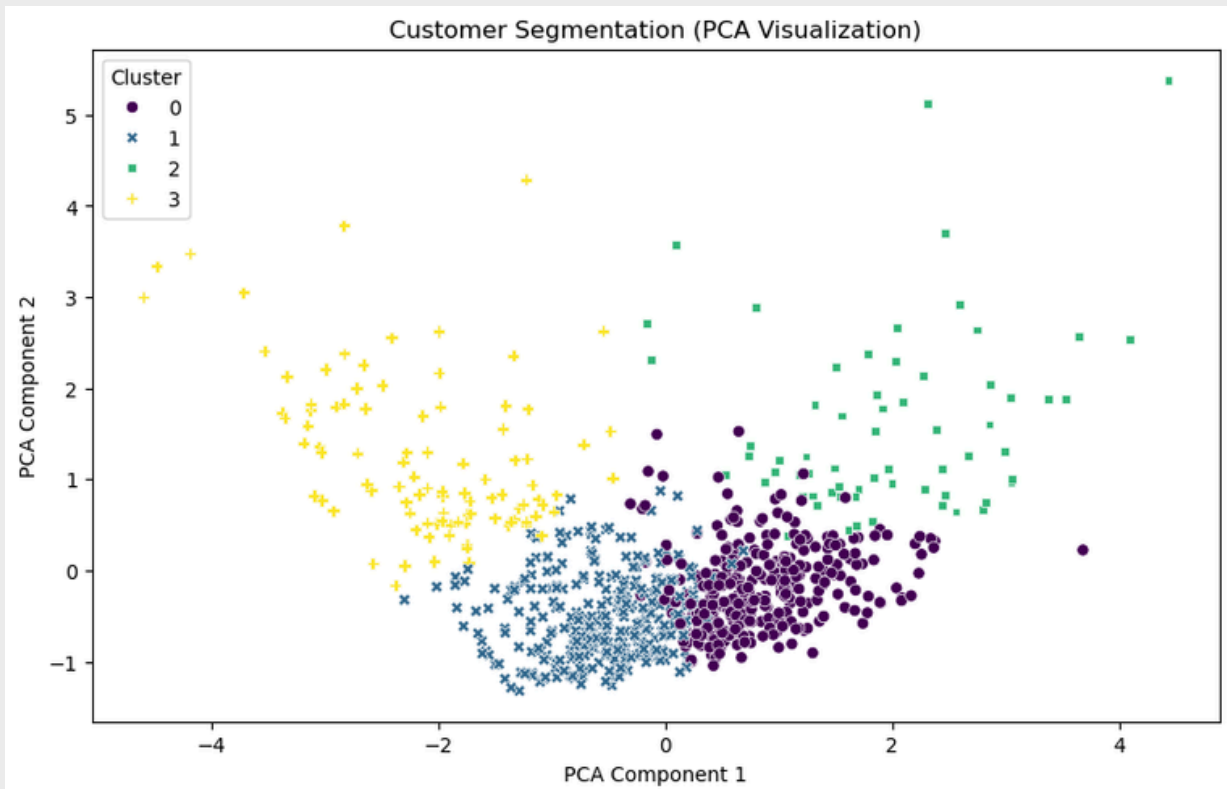
What the results mean for marketing:

Four segments emerged with meaningfully different profiles. The decision implication of each is different:

Segment	Recency	Frequency	Avg Spend	Avg Profit	Size	Marketing implication
Champions (Cluster 0)	72 days	8.5 orders	\$3,308	\$350	300 (38%)	Retain with non-discount perks
Potential Loyalists (Cluster 1)	102 days	4.7 orders	\$1,672	\$183	333 (42%)	Nurture toward higher frequency
Loyal High-Spenders (Cluster 2)	124 days	8.3 orders	\$9,480	\$1,676	64 (8%)	Protect at all costs — highest CLV
At-Risk Customers (Cluster 3)	559 days	3.7 orders	\$1,470	\$139	96 (12%)	Reactivate or accept churn

Cluster 2 (Loyal High-Spenders) is the most commercially critical: just 64 customers (8% of the base) but averaging \$1,676 profit each — nearly five times the next best group. Cluster 3 shows a recency of 559 days, meaning these customers have been absent for over 18 months on average.

4.1. Segmentation (RFM Feature Construction & K-Means Clustering)



The two axes are not raw variables — they are mathematical summaries of the five RFM features compressed into two dimensions for visualisation. Component 1 captures the largest source of variation across customers; Component 2 the next most important.

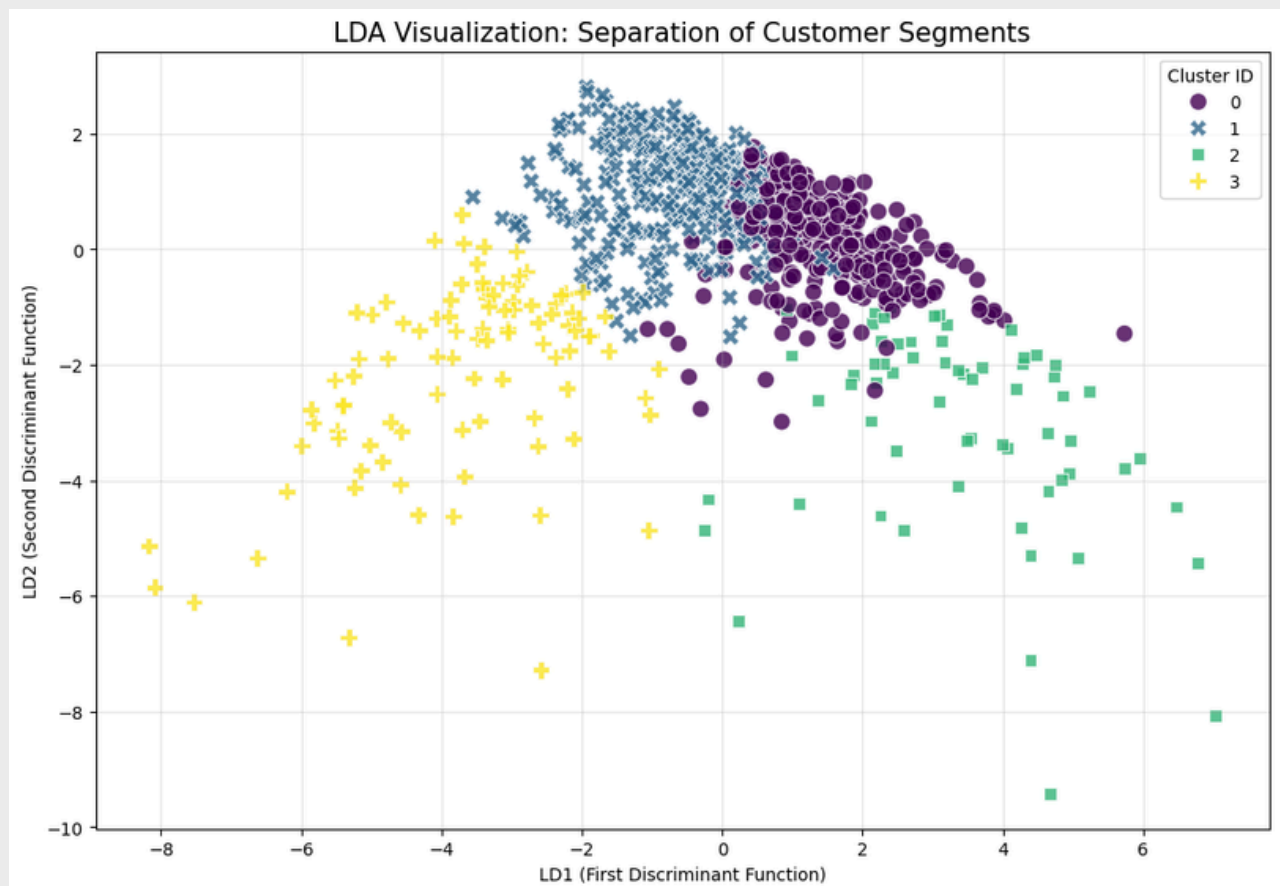
What the plot reveals: Each dot is one customer, coloured by K-Means cluster. Cluster 2 (Loyal High-Spenders) sits clearly apart, reflecting their outlier spend. Cluster 3 (At-Risk) is pulled in the opposite direction, driven by long recency and low frequency. Some overlap between Clusters 0 and 1 is expected — they differ more in spending depth than in the dimensions PCA prioritises.

4.2. LDA (Linear Discriminant Analysis)

Objective: To validate whether the K-Means customer segments are statistically meaningful, Linear Discriminant Analysis (LDA) was applied. LDA predicts which cluster a customer belongs to using five behavioral features: Recency, Frequency, Monetary value, Total Profit, and average Discount rate. Unlike K-Means, which groups customers by distance, LDA tests whether those groups are genuinely separable by linear boundaries — confirming the segments are real behavioral archetypes rather than mathematical artifacts.

4.2.1. LDA Visualization: Separation of Customer Segments

The plot below projects all 793 customers onto the two primary LDA discriminant axes (LD1 & LD2). Each point represents 1 customer, coloured by their K-Means cluster assignment.



The visualization confirms that the four segments form clearly distinguishable clouds in discriminant space. Cluster 2 (Loyal High-Spenders) sits completely isolated from the other three groups, reflecting their extreme spending outlier status. Cluster 3 (At-Risk) is pulled to the far left, driven by their long recency and low frequency scores. Clusters 0 and 1 show some spatial overlap — this is expected, as Champions and Potential Loyalists differ more in spending depth than in the dimensions LD1 and LD2 prioritise.

Why LDA was chosen: LDA was selected over alternatives such as logistic regression or random forest because it is specifically designed to find linear combinations of features that maximally separate known classes. Given that the K-Means clusters were built from continuous behavioral features, LDA is the natural validation tool — it uses the same feature space to test whether the boundaries are statistically meaningful.

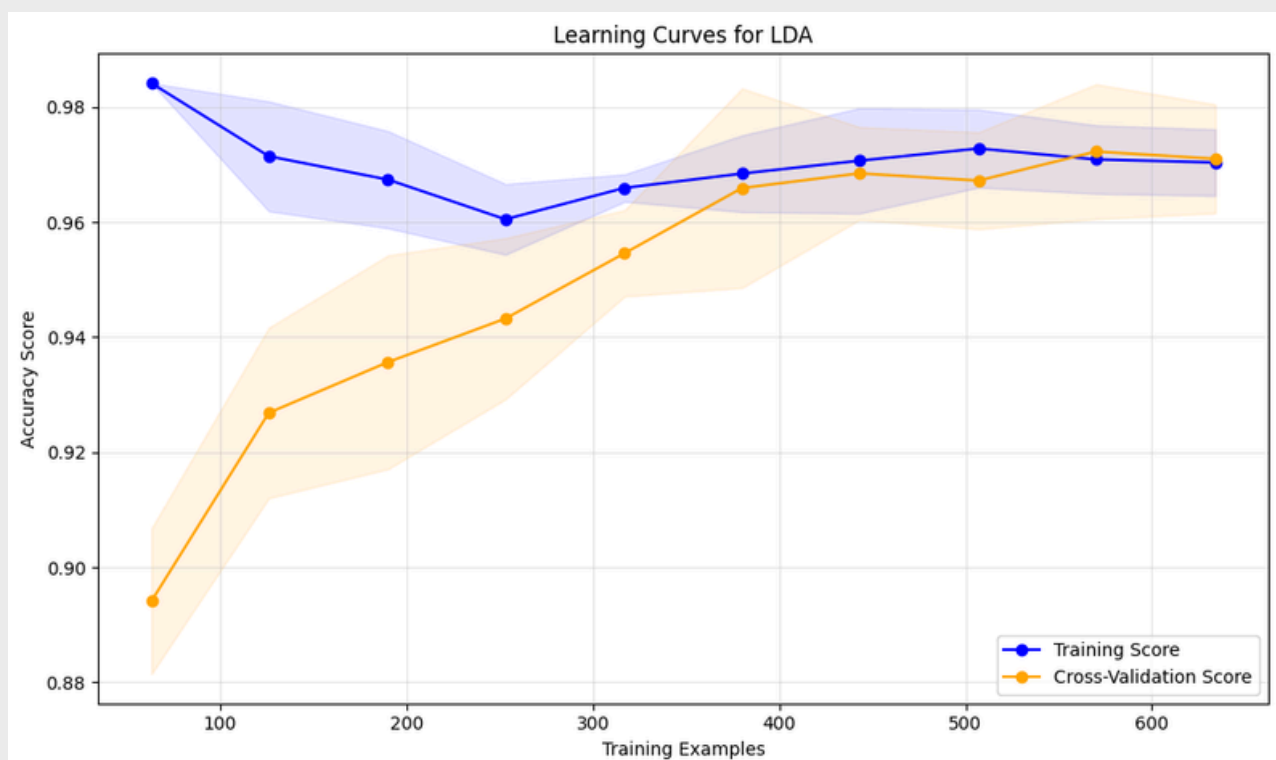
Data Preparation: An RFM profile was built for each of the 793 unique customers. Recency, Frequency, Monetary, and Total Profit were computed at the customer level. Discount was derived as the average discount rate per customer across all their transactions, joined from the order-level dataset. The five features were then combined into a single customer-level matrix before splitting.

Train/Test Split: The 793 customers were split into a training set of 555 (70%) and a test set of 238 (30%), with random state fixed at 42 for reproducibility. The four K-Means cluster labels served as the target variable.

Cross-Validation: 5-fold stratified cross-validation was applied to confirm model stability. Fold scores were: 0.956, 0.950, 0.950, 0.981, and 0.987, yielding a **mean CV accuracy of 96.5% (+/-3.2%)** — closely aligned with the test set result, confirming no overfitting.

4.2.2. Learning Curves for LDA

To confirm that the model is stable and not simply memorising the training data, learning curves were plotted by gradually increasing the training sample size from 10% to 100% and measuring both training accuracy and 5-fold cross-validation accuracy at each point.

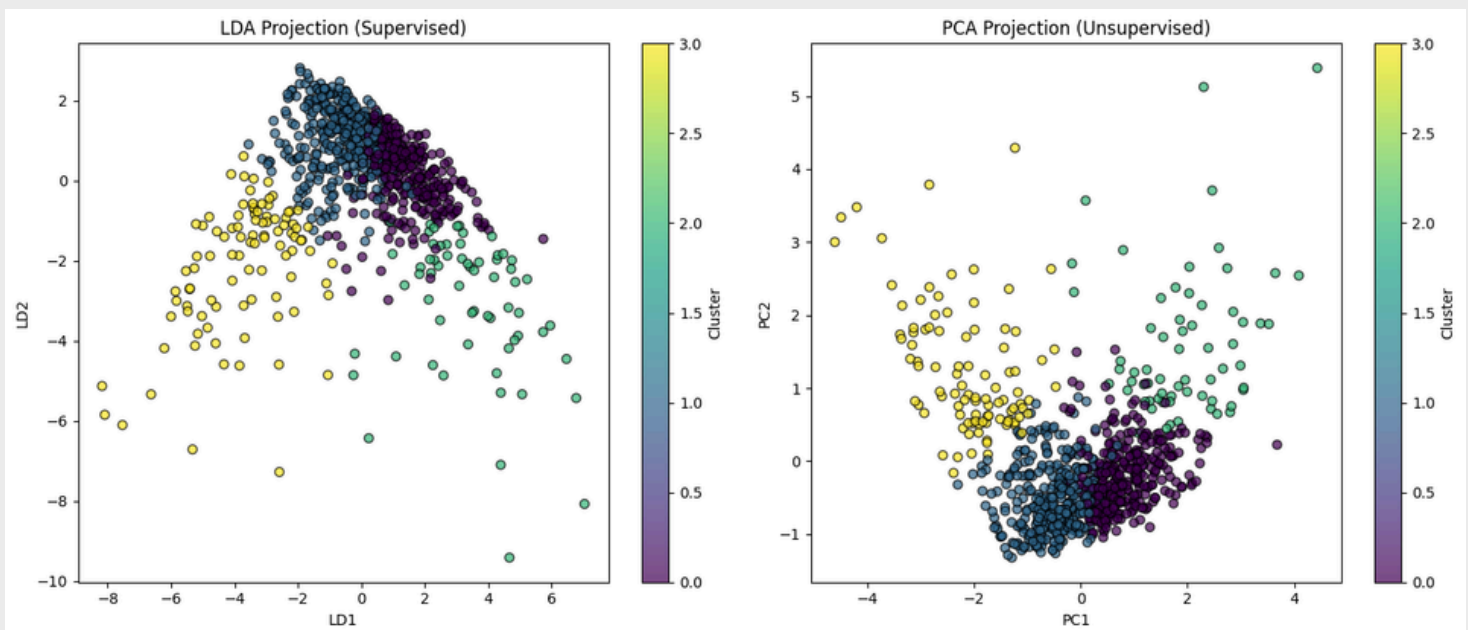


The **training score (blue line)** and **cross-validation score (orange line)** converge closely as the sample size grows, stabilising above 96% accuracy. The narrow gap between the two lines confirms the model is **not overfitting** — it generalises well to unseen customers. The slight decline in training accuracy as more data is added is normal and healthy behavior for LDA, indicating the model is learning genuine patterns rather than noise.

Discriminant Functions: LDA produced three discriminant functions. **LD1 explains 62.3% of between-class variance, LD2 explains 28.8%, and LD3 explains 8.9%.** Together, LD1 and LD2 capture **91.1% of the total separating power**, which is why the 2D projection is sufficient to visualise the segment separation clearly.

4.2.3. LDA Projection (Supervised)" + "PCA Projection (Unsupervised)

To further validate that the segments are meaningful, the LDA projection (supervised — class-aware) is compared directly against a PCA projection (unsupervised — class-blind) of the same data.



The contrast is striking. In the LDA projection (left), the four clusters form tight, well-separated groupings — particularly Cluster 2 and Cluster 3, which are pulled to opposite extremes of the discriminant space. In the PCA projection (right), the same customers appear largely overlapping and indistinguishable, because PCA maximises total variance without any awareness of class boundaries. This comparison is strong evidence that the four customer segments are not random mathematical groupings — they represent genuinely distinct behavioral patterns that LDA can detect and separate with 94.5% accuracy.

4.3. Customer Lifetime Value (CLV)

Objective: The objective of this analysis is to identify which customers should be prioritized for resource allocation using a data-driven Customer Lifetime Value (CLV) framework.

Given limited marketing and retention resources, it is not efficient to target all customers equally. Therefore, this analysis focuses on customer selection, rather than simple customer description.

4.3.1. Process

This analysis is implemented in Python using a structured workflow combining:

1.Data Preparation

Transform transactional data into customer-level datasets

2.Rolling Snapshot Construction

Generate multiple time-based training samples

3.Feature Engineering

Extract behavioral and value-related features

4.Model Development

Predict future customer value and high-value probability

5.Customer Ranking & Segmentation

Build a prioritization framework

6.Business Interpretation

Translate model outputs into actionable strategies

4.3.2. Analytical Approach

4.3.2.1. Rolling Snapshot Design

Customer behavior is captured at multiple time points (rolling snapshots), allowing the model to learn from temporal patterns rather than a single static view.

This significantly increases the number of training samples and improves model robustness.

4.3.2.2. Feature Engineering

Customer-level features are derived from transactional data, including:

- **Recency** (days since last purchase)
- **Frequency** (number of orders)
- **Monetary value** (total sales, profit)
- **Product diversity** (number of unique products/categories)
- **Behavioral patterns** (discount usage, profitability, purchase consistency)

These features are designed to capture both value and behavior dynamics.

4.3.2.3. Modeling Framework

A two-part modeling approach is used:

1. Regression model

Predicts the magnitude of future profit

2. Classification model

Predicts the probability of being a high-value customer (top 20%)

Model performance

The optimized model demonstrates strong business-oriented performance for customer selection.

While the regression results suggest moderate error in exact profit prediction, the ranking and targeting metrics are much stronger. In particular, the model captures 48.35% of future value within the top 20% of selected customers, and the top 10% shortlist reaches 83.54% precision. This confirms that the model is well suited for prioritization, targeting, and CRM resource allocation.

	Metric	Value
0	MAE	141.6847
1	RMSE	437.9770
2	Spearman	0.4764
3	ROC_AUC	0.7393
4	PR_AUC	0.6137
5	Capture_at_20pct	0.4835
6	Precision_at_20pct	0.6433
7	Capture_at_10pct	0.3616
8	Precision_at_10pct	0.8354

4.3.2.4. Customer Ranking & Prioritization

Customers are ranked based on a selection score, combining:

- Predicted future profit
- Probability of being high-value

Customers are then segmented into:

- Priority A (Top targets)
- Priority B (Mid-value potential)
- Priority C (Low priority)

4.4. Category New-product Sales Predict

Datasets: US Superstore

Objective: This data analysis and forecasting project aims to address the challenges of sales forecasting during the initial launch phase of new products. The core business objectives include:

1. Establishing a benchmark for simulation
2. Quantifying future sales trends
3. Capturing seasonal market cycles
4. Empowering key business decision-making

4.4.1. Process

Conduct the analysis using a standard data analysis workflow:

- Data Reading and Loading
- Data cleaning and pre-processing
- Data filtering and resampling: The user selects a 'reference subcategory' (for example, 'Chairs' in this test) to serve as the historical benchmark data for the new product. And resample the data in this category on a monthly basis (MS, i.e. at the start of each month) and calculate the total monthly sales.
- Model Building and Forecasting
- Result Output and Visualisation

4.4.2. Methodology

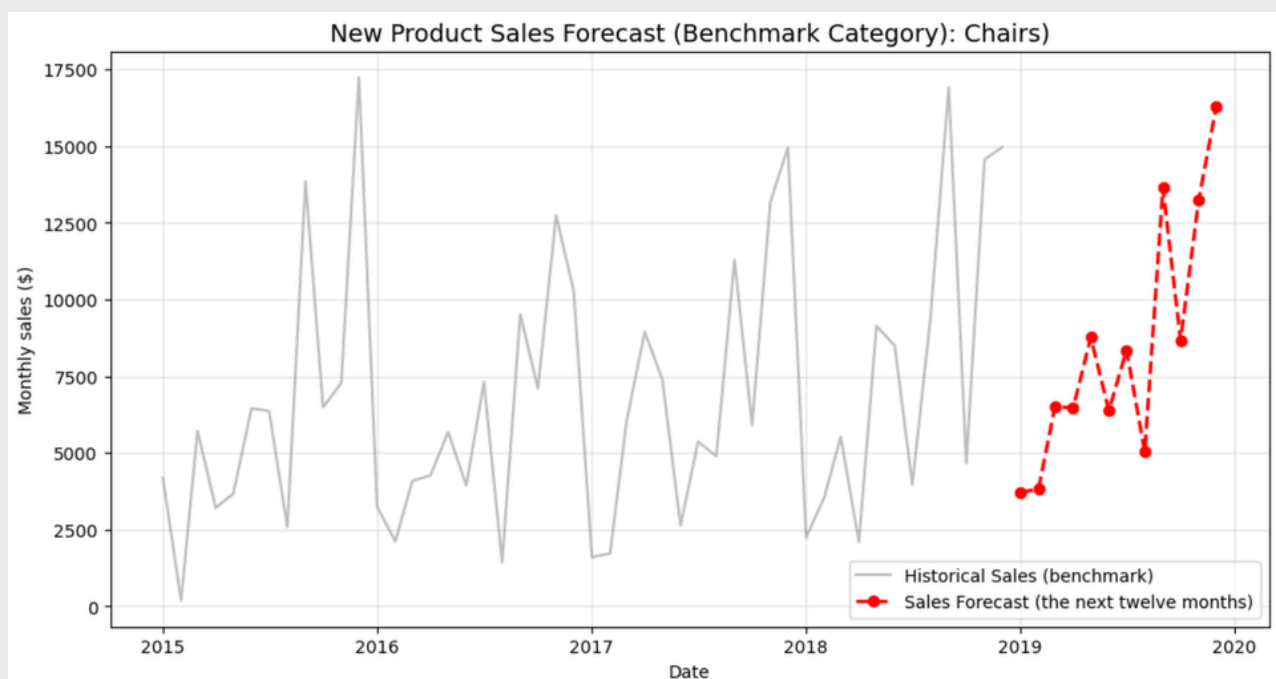
This project primarily employs the Holt-Winters exponential smoothing method from time series analysis :

- The model utilises additive trend (trend='add') and additive seasonal (seasonal='add') components
- Damping trends: It assumes that, over time, the rate of growth or decline in a trend will gradually level off, thereby avoiding extreme or unrealistic over-projections of future sales.
- Exception handling mechanism: If there is insufficient historical data for a particular benchmark category to support complex seasonal analysis, the model will automatically downgrade and use a simple exponential smoothing model—which incorporates only the trend—to make forecasts.

4.4.3. Results

Visualisation results: The line chart successfully illustrates historical sales figures (grey solid line) alongside sales forecasts for the next 12 months (red dotted line). The graph shows that the forecasting model has effectively captured the seasonal fluctuations in the historical data (such as the typical year-end sales peak shown in the figure).

Conclusion: If the market performance of the new product is similar to that of Chairs, the company should make advance preparations for stock levels and supply chain planning by the end of the first quarter and in the second half of the year to cope with the pronounced seasonal sales peak.



4.5. Sales Forecasting

Datasets: Adidas US Sales Data

Objective:

- **Accurate forecasting of sales trends:** Forecasting future product sales volumes for specific time periods based on Adidas' historical sales data.
- **Identifying key drivers:** identifying the core factors that influence sales to guide the allocation of marketing resources.
- **Optimising supply chains and inventory:** By providing accurate sales forecasts, we help businesses plan production and warehousing in advance, thereby reducing the costs associated with excess stock whilst preventing stock-outs of best-selling items.

4.5.1. Process

Conduct the analysis using a standard data analysis workflow:

- **Data loading and pre-processing:** Read the historical sales dataset from Adidas, clean missing and outlier values, and convert the date column to a standard time format.
- **Time-series feature engineering:** This is key to time-series forecasting. It involves extracting multi-dimensional features from dates, such as the year, month, day, day of the week, quarter, and whether it is a weekend or a public holiday.
- **Data split:** To prevent 'future data leaks', the dataset is strictly divided into a training set (earlier time period) and a test set (later time period) in chronological order.
- **Model creation and training:** Define an XGBoost regression model, set key hyperparameters (such as `learning_rate`, `max_depth` and `n_estimators`), and fit the model using the training data.
- **Model evaluation and analysis:** Generate predictions on the test set, calculate error metrics, and plot comparison charts of actual and predicted values, as well as feature importance charts.

4.5.2. Methodology

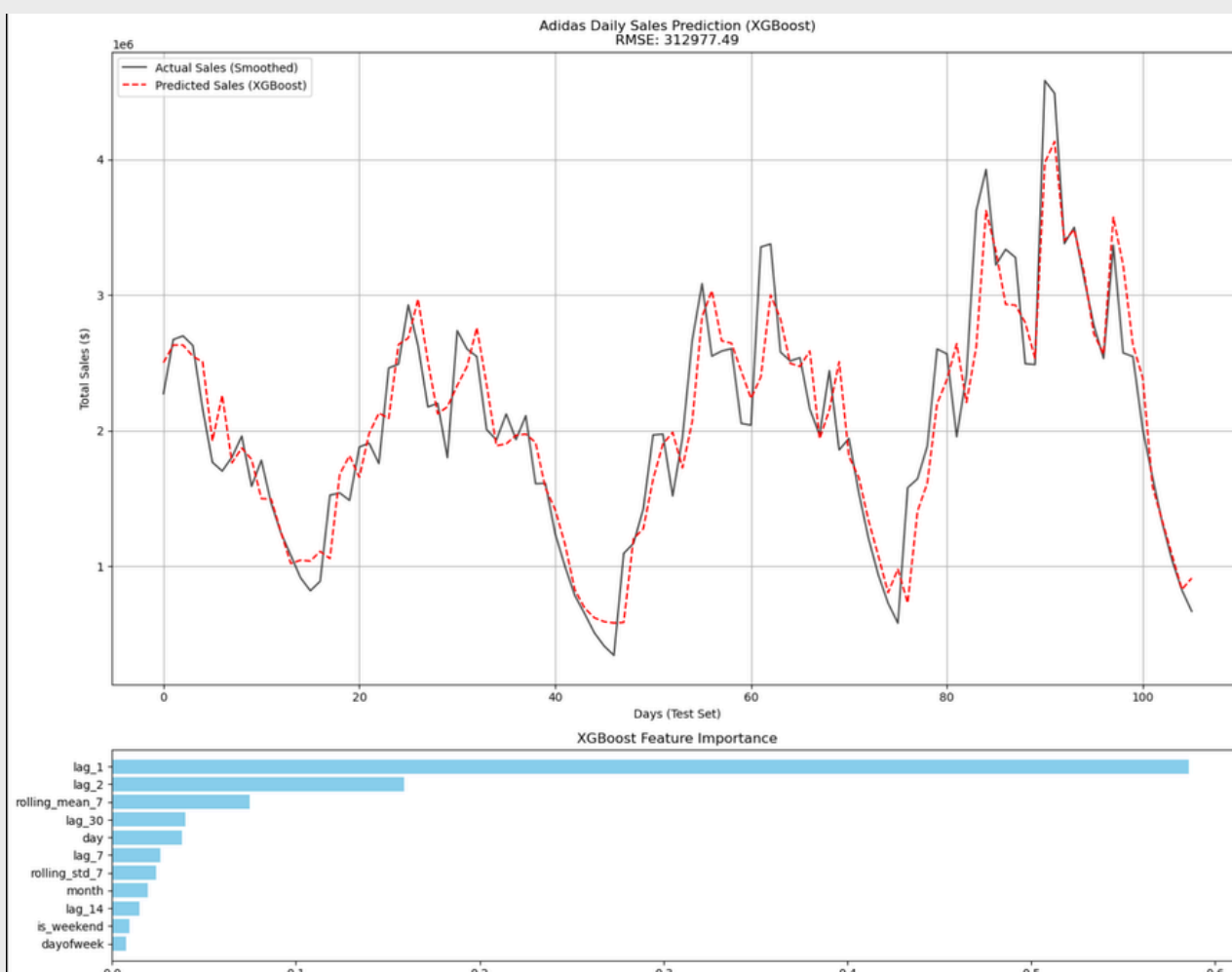
- The core algorithm used in this project is XGBoost, a decision tree-based ensemble machine learning algorithm that performs exceptionally well when processing tabular data.

4.5.3. Results

Model fitting performance: By plotting a line graph comparing predicted sales against actual sales on the test set, it is clear that the model has successfully captured the fluctuating trends and peaks in Adidas's sales data (such as the sharp spikes in sales during seasonal changes or specific shopping festivals).

Error quantification: The model outputs key evaluation metrics (such as RMSE or MAE), demonstrating that the model possesses strong predictive capabilities within an acceptable margin of error.

Feature Importance: The model's output of feature weight rankings reveals which factors have the greatest influence on Adidas's sales (for example, 'specific month' may determine sales of seasonal footwear and apparel, whilst 'whether a promotion is included' may be the most critical feature driving sales peaks).

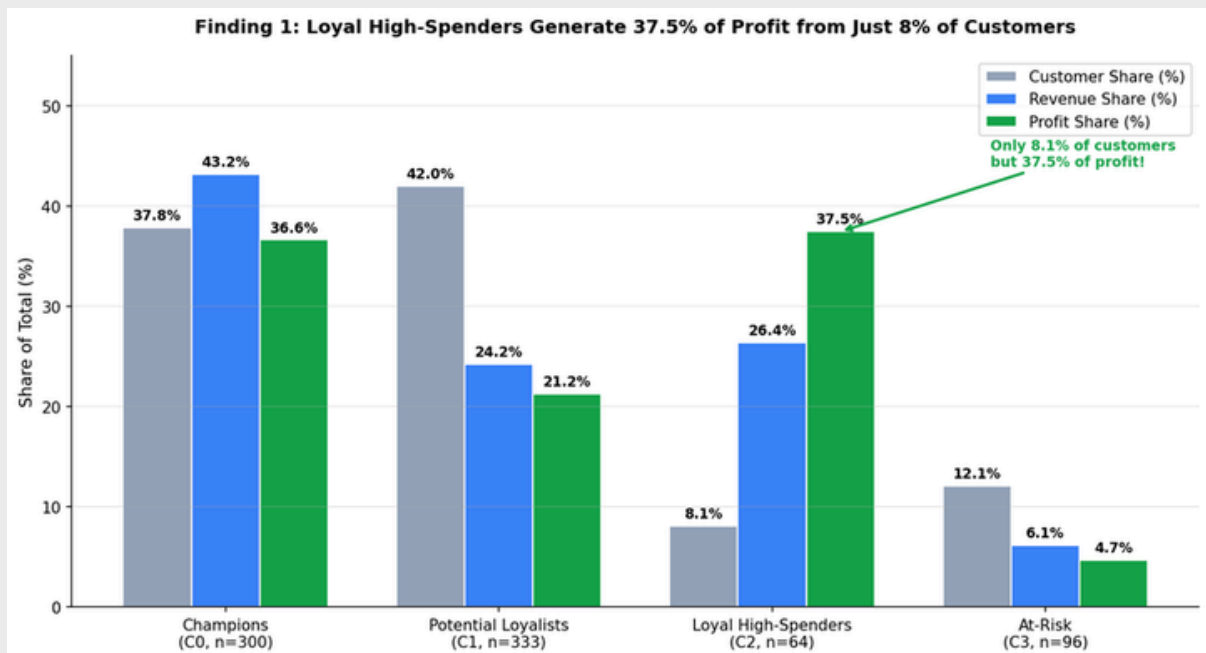


5. Key Results and Findings

5.1. Segmentation (RFM Feature Construction & K-Means Clustering)

5.1.1. Loyal High-Spenders Generate 37.5% of Profit from Just 8% of Customers

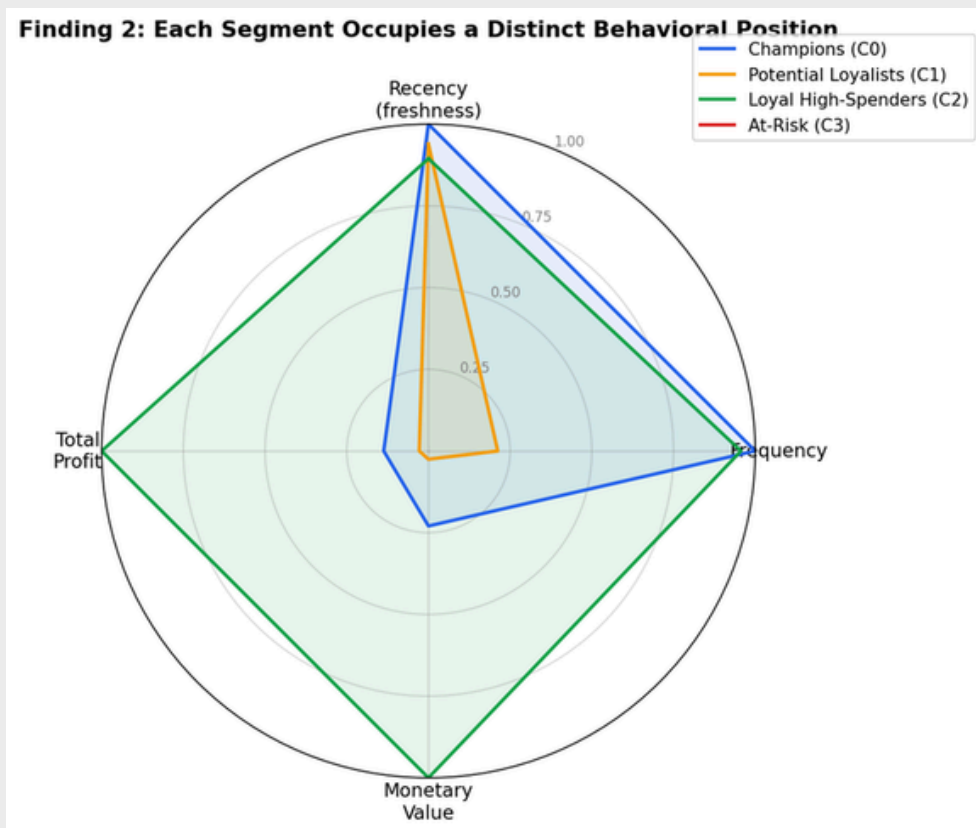
The most striking result of the segmentation is the extreme commercial imbalance across clusters. Loyal High-Spenders (Cluster 2) account for only 8.1% of the customer base (64 customers), yet generate 26.4% of total revenue and 37.5% of total profit. By contrast, Potential Loyalists (Cluster 1) — the largest group at 42% of customers — contribute only 21.2% of profit. Champions (Cluster 0) are both large (38%) and profitable (36.6%), making them the backbone of day-to-day revenue. At-Risk customers (Cluster 3) are effectively disengaged, contributing a negligible 4.7% of profit despite representing 12% of the base.



5.1.2. Each Segment Occupies a Distinct Behavioral Position

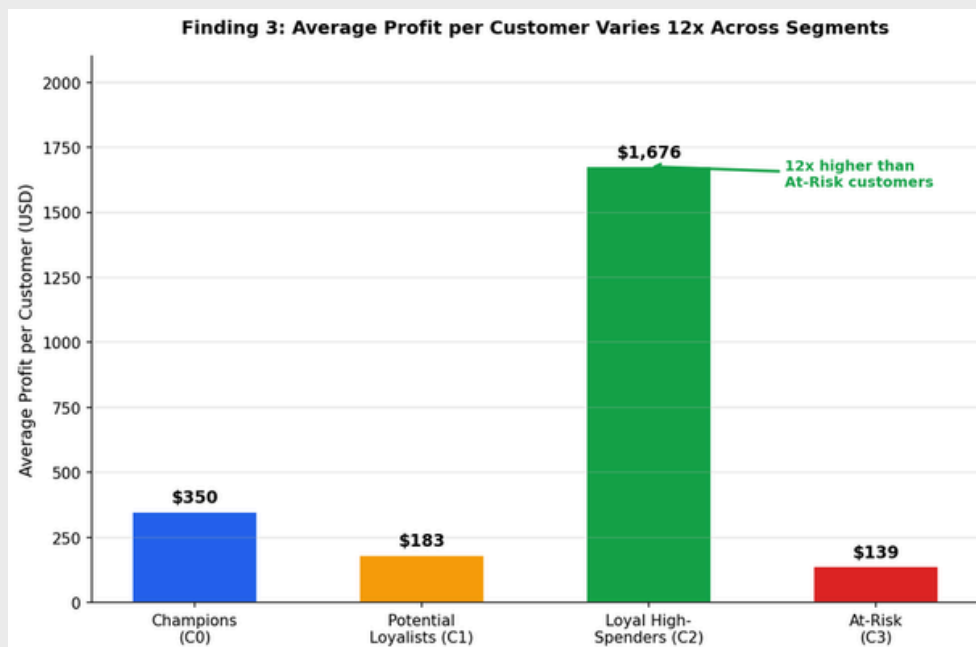
The radar chart below plots each cluster across four normalised dimensions: Recency (freshness of last purchase), Frequency, Monetary value, and Total Profit. The four clusters form clearly non-overlapping shapes, confirming that each segment represents a genuine and actionable behavioral archetype rather than an arbitrary partition:

- Champions (C0): Strong on Recency and Frequency — active and engaged buyers, solid but not exceptional spenders.
- Loyal High-Spenders (C2): Dominant on Monetary and Profit — the highest-value customers by spend and margin, though slightly less frequent than Champions.
- Potential Loyalists (C1): Moderate across all dimensions — a broad middle group with room to grow in both frequency and spend.
- At-Risk (C3): Near-zero on Recency — these customers have effectively stopped buying. Their Frequency and Monetary scores are also the weakest.



5.1.3. Profit per Customer Varies 12x Across Segments

When normalised to a per-customer basis, the profit gap between segments becomes even sharper. Loyal High-Spenders average \$1,676 profit per customer — 4.8x more than Champions (\$350), 9.2x more than Potential Loyalists (\$183), and 12x more than At-Risk customers (\$139). This finding has direct budget allocation implications: acquiring or retaining one Loyal High-Spender is worth the same as retaining 12 At-Risk customers in profit terms. Marketing investment should be strongly weighted toward protecting Cluster 2 and converting Cluster 0 (Champions) into Cluster 2 through cross-sell and upsell programmes.

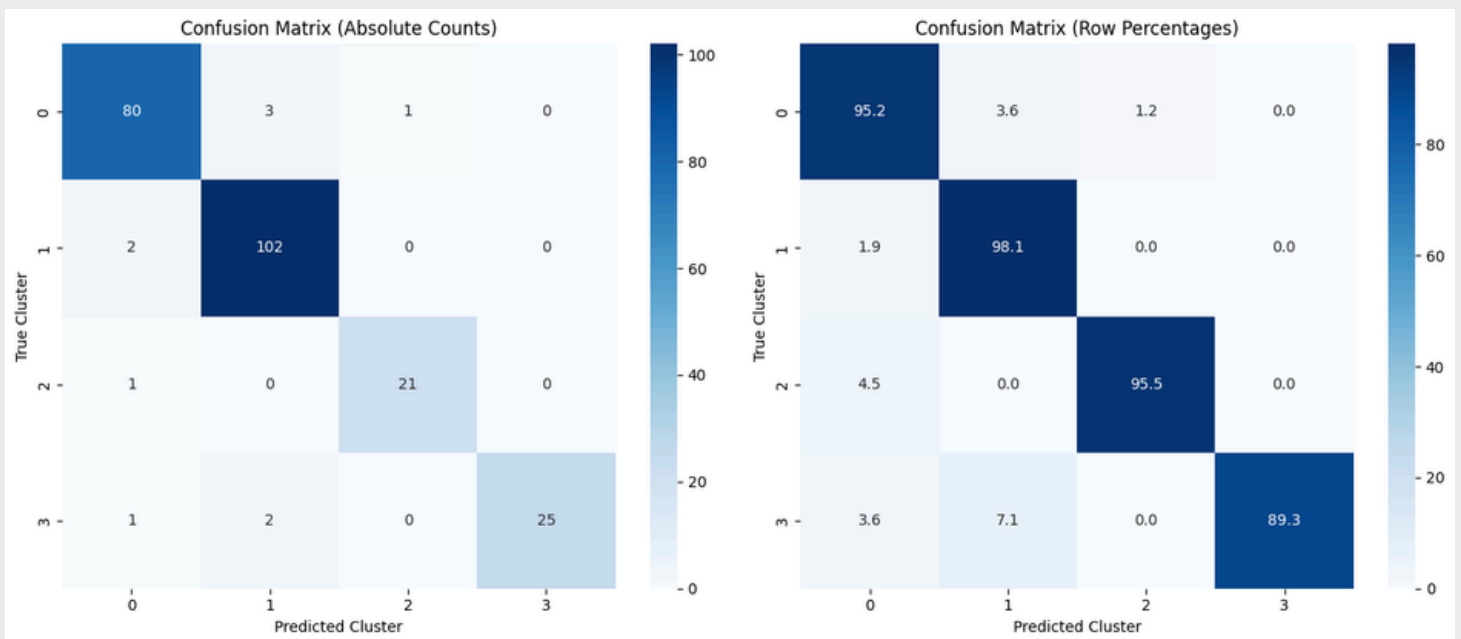


5.2. LDA (Linear Discriminant Analysis) Results

5.2.1. Confusion Matrix: 94.5% Test Accuracy

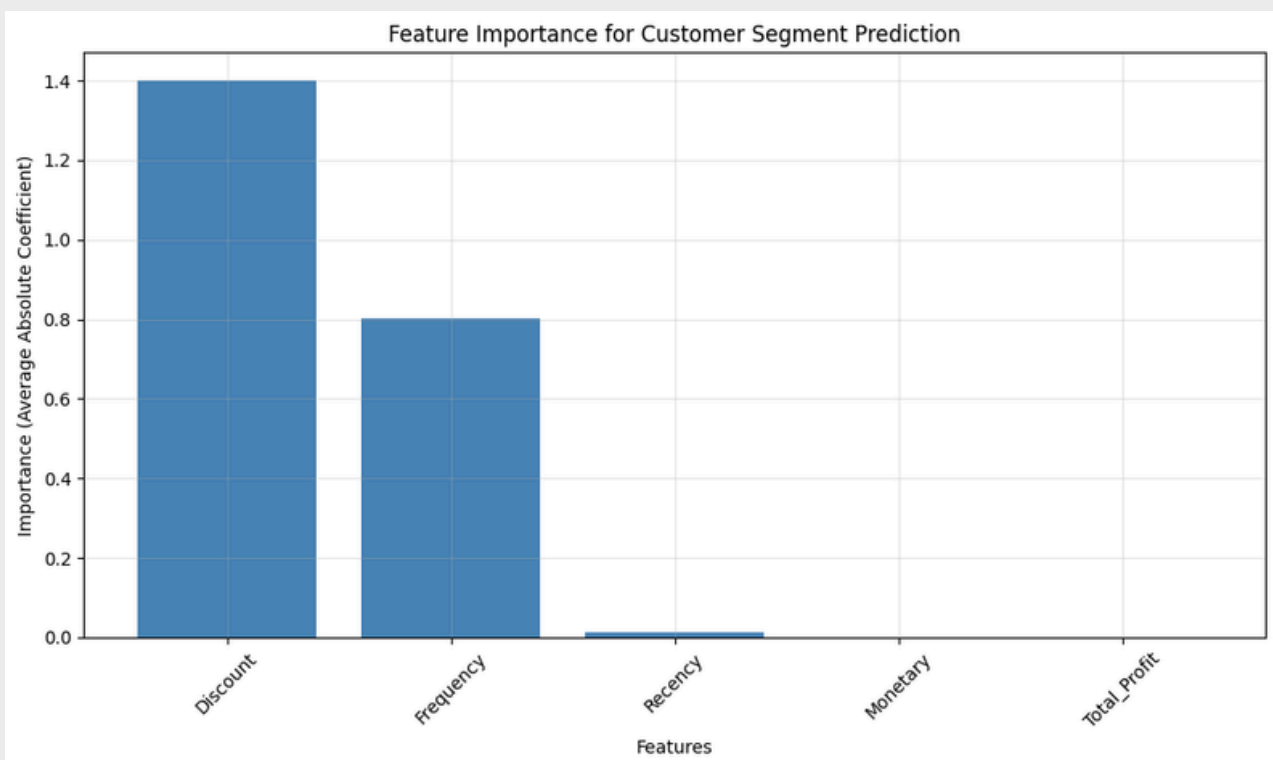
The LDA model achieved **94.5% accuracy on the test set** (238 customers), correctly classifying 225 out of 238 customers. Only 13 customers were misclassified — a **5.5% error rate**. Per-cluster results are as follows:

Cluster	Segment	Precision	Recall	F1-Score	Support
0	Champions	93.00%	95.20%	94.10%	84
1	Potential Loyalists	94.50%	99.00%	96.70%	104
2	Loyal High-Spenders	94.70%	81.80%	87.80%	22
3	At-Risk Customers	100.00%	85.70%	92.30%	28



5.2.2. Feature Importance: Discount and Frequency Drive Segment Differences

LDA coefficients reveal that **Discount (importance score: 1.541)** and **Frequency (0.798)** are overwhelmingly the strongest predictors of segment membership. The remaining three features — Recency (0.012), Monetary (0.001), and Total Profit (0.001) — contribute almost nothing to separating the segments in discriminant space. This is a critical finding: it means customers are not primarily separated by how much they spend, but by **how often they buy and whether they rely on discounts to do so**. Champions buy frequently with minimal discounts; At-Risk customers do the opposite.



5.2.3. Operational Implication: Real-Time Segment Assignment

With 94.5% test accuracy and 96.5% cross-validated accuracy, the LDA model is robust enough for real-world CRM deployment. Because Discount and Frequency are the two dominant predictors, a customer's segment can be reliably determined after just a small number of transactions — without waiting months for a full behavioral history to accumulate. A new customer making a high-value first purchase at zero discount can be **immediately flagged as a potential Champion** and enrolled in a loyalty nurture track, preventing the costly mistake of sending them an unnecessary discount.

5.2.4. At-Risk and Loyal High-Spender Misclassification Risk

Two clusters show lower recall and warrant attention. **Cluster 2 (Loyal High-Spenders)** had the lowest recall at **81.8%** — meaning 4 out of 22 Loyal High-Spenders were misclassified, likely confused with Champions due to their similar high frequency. This is the highest-risk misclassification commercially: a Loyal High-Spender incorrectly treated as a **Champion** may not receive the premium upsell and cross-sell communications they warrant.

Cluster 3 (At-Risk) had **85.7% recall**, with 4 customers misclassified — 1 confused with Champions and 3 confused with Potential Loyalists. To mitigate this, it is recommended that the CRM team apply a **secondary recency filter (no purchase in the last 180 days)** before deploying any At-Risk reactivation campaign, as an additional safeguard on top of the LDA classification.

Graph: Reference the **confusion matrix** above.

5.2.5. Conclusion

The LDA analysis delivers three clear takeaways for the business.

- **First**, the four customer segments are statistically real, supported by **94.5% test accuracy** and **96.5% cross-validated accuracy** using only five features.
- **Second**, discount behavior is the most powerful differentiator between customers, with an importance score of **1.541**, far exceeding all other variables.
- **Third**, the model is ready for real-world deployment, with two safeguards required:
 - a. Apply additional verification for Loyal High-Spenders
 - b. Use a 180-day recency filter for At-Risk customers

5.3. Customer Lifetime Value (CLV)

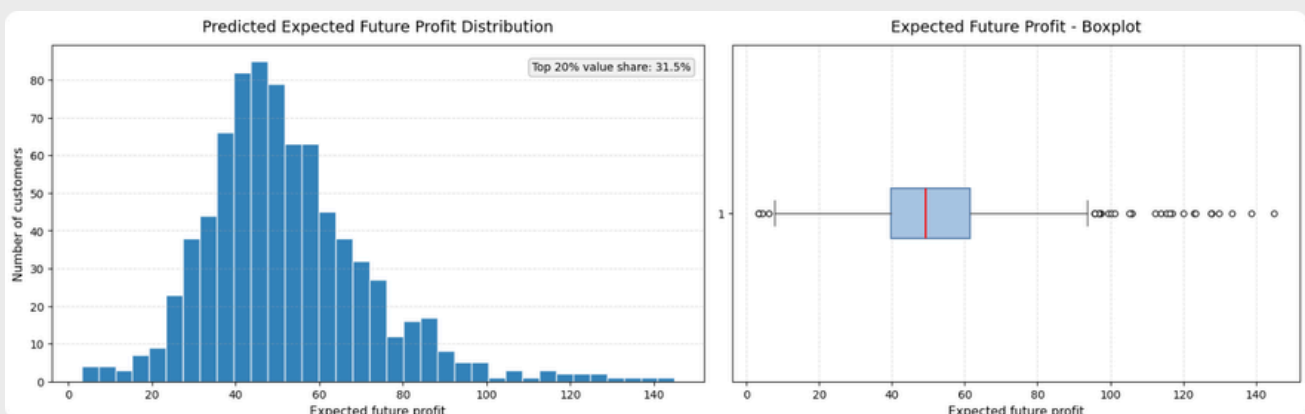
5.3.1. Customer value is highly concentrated

The Pareto chart shows that the top 20% of customers account for about 65.2% of historical positive profit. This is strong evidence against equal treatment: the base is economically uneven, so customer-selection logic is necessary.



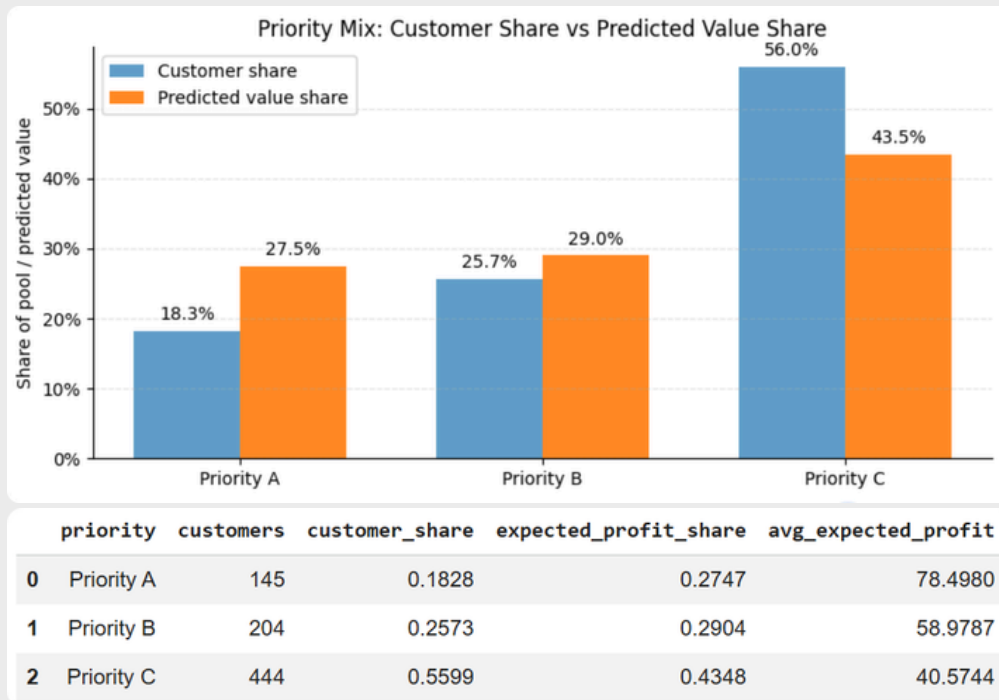
5.3.2. Predicted future value is still long-tailed

The expected future profit distribution remains skewed in the current pool. The average expected future profit is 52.2, while the median is 49.6. The top 20% of customers capture about 31.5% of total predicted value, which again supports selective allocation rather than uniform investment.



5.3.3. Priority A is small but economically important

Priority A contains 145 customers (18.9% of the pool), yet it contributes 27.4% of predicted future value. This means the highest-priority layer over-indexes on value relative to its size



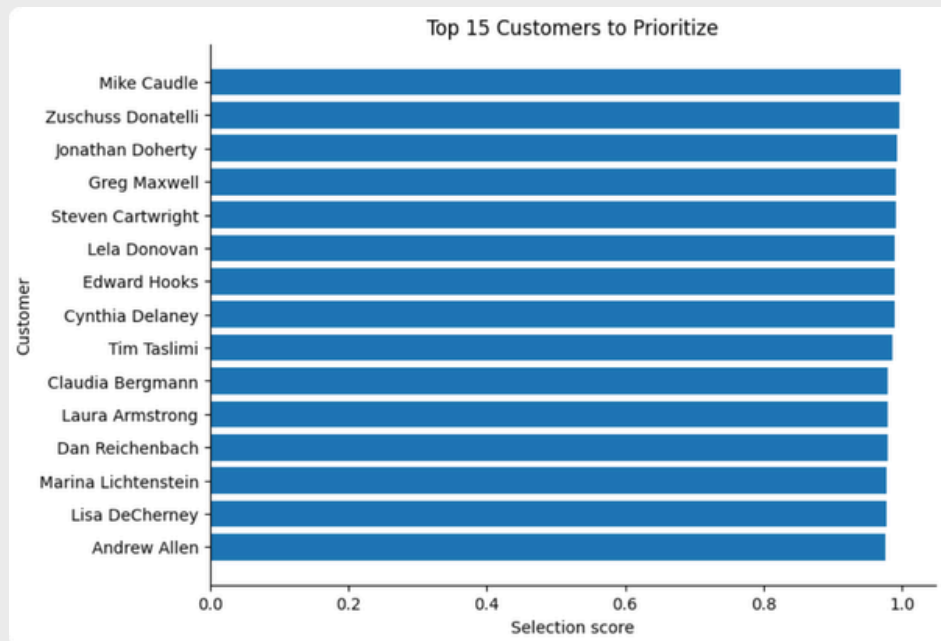
5.3.4. Who should we prioritize?

The upper-right quadrant is the most attractive zone because it combines stronger expected value with a stronger likelihood of belonging to the high-value group. Those customers should receive the first layer of high-touch attention.



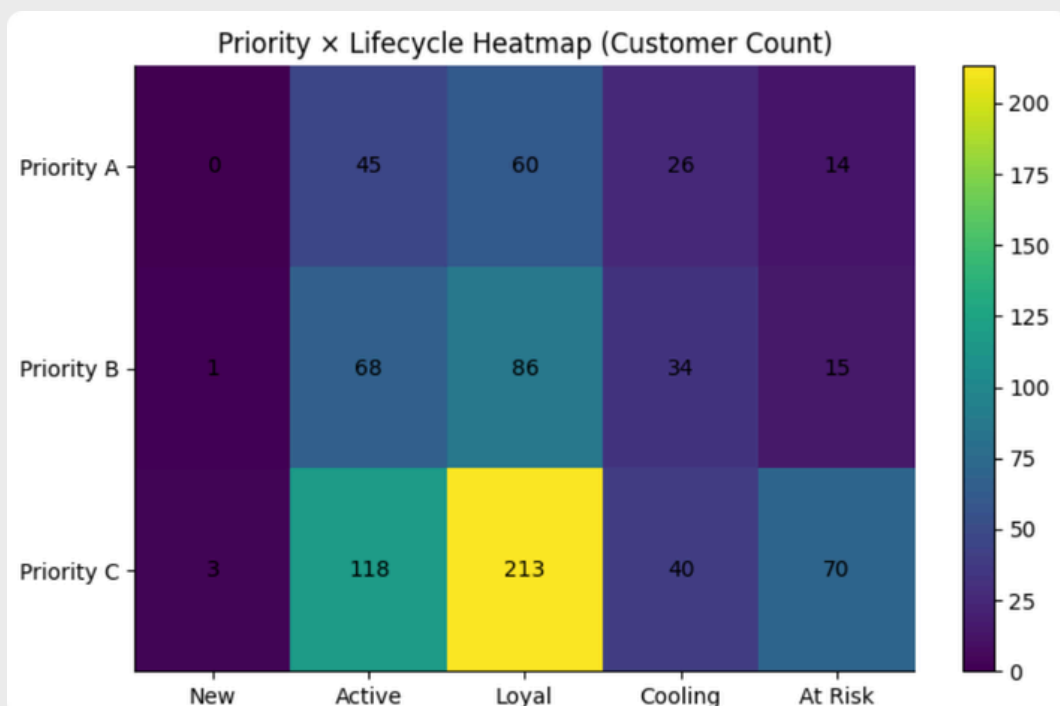
5.3.4.1 Top target list

The top target list is the first working queue for sales and CRM teams.



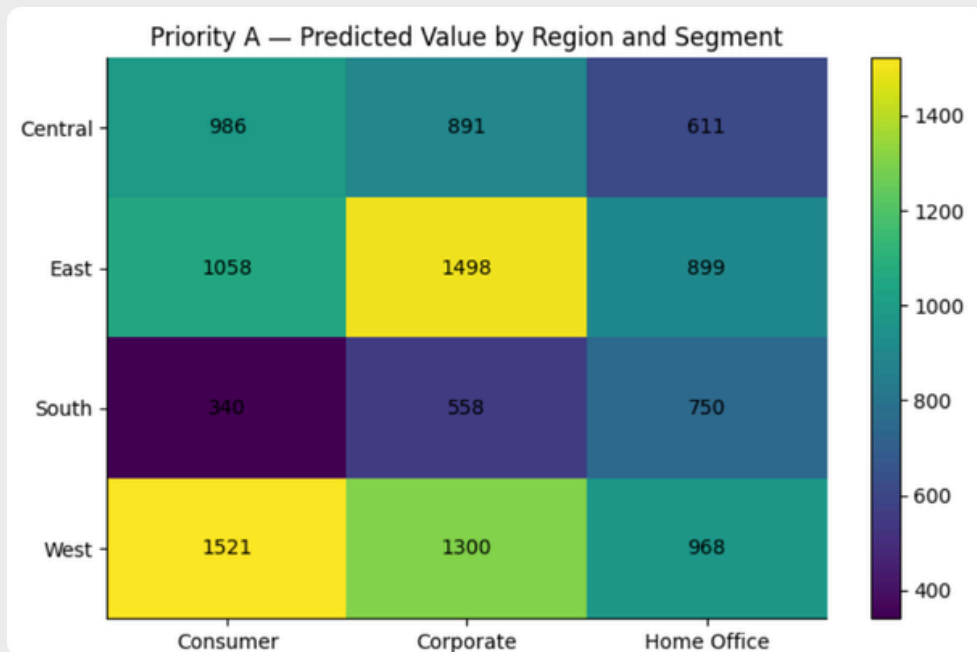
5.3.5. What action should we take by customer type?

Within Priority A, the largest operational groups are 60 Loyal customers (retain + upsell / cross sell), 45 Active/New customers (accelerate the next order), and 40 Cooling or At Risk customers (win-back + tailored offer + sales outreach). This is why action should depend on both value and lifecycle, not value alone.



5.3.6. What action should we take by customer type?

Priority A value is not evenly distributed. By region, West contributes about 35.4% of Priority A value and East about 28.3%. By segment, Corporate contributes about 37.6% and Consumer about 35.0% of Priority A value. This means that after identifying Priority A customers, the next layer of targeting can be refined by region and segment to improve coverage efficiency.

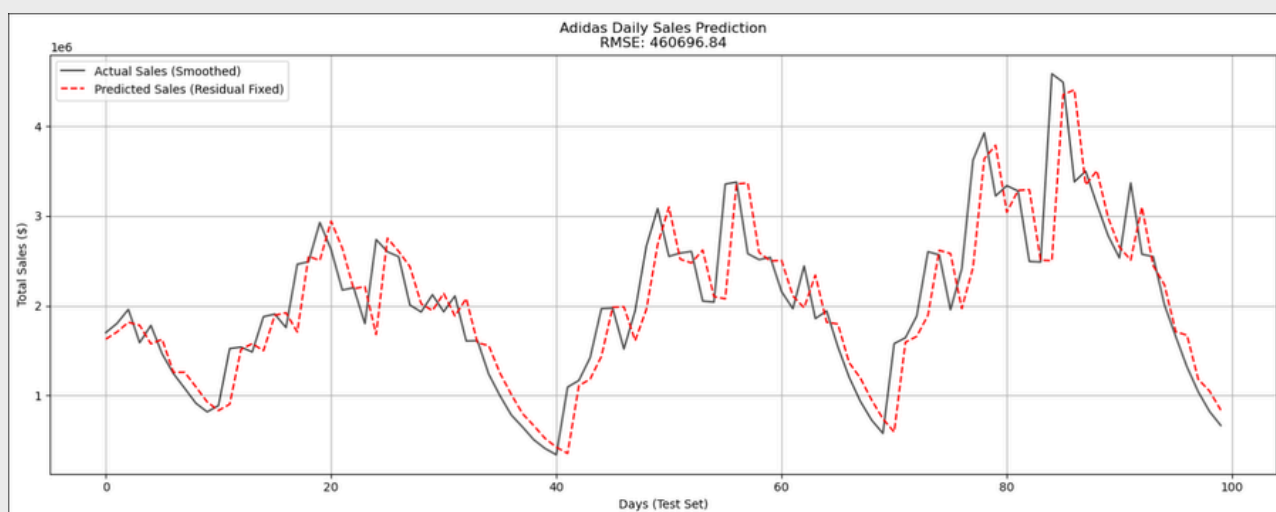
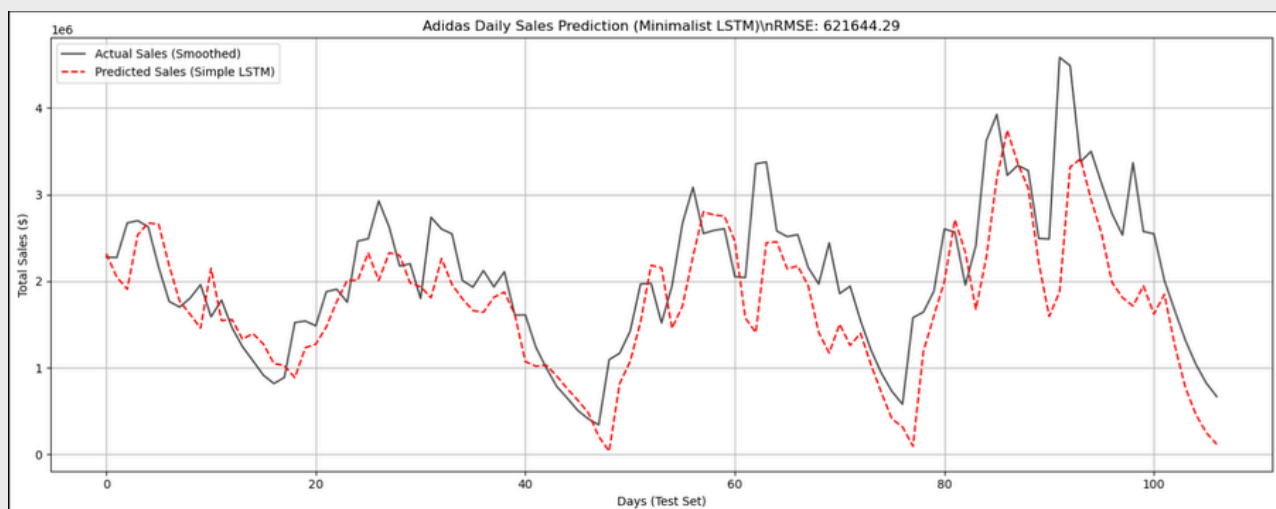


5.3.7. Conclusion

In summary, the optimized CLV framework supports a clear business recommendation: prioritize the customer base instead of treating it uniformly. The model is validated for selection and ranking, the current pool shows clear value concentration, and the resulting Priority A/B/C structure can be translated into concrete action for sales, CRM, and retention teams. The most practical next step is to operationalize the top-target list and measure campaign uplift by priority tier.

5.5. About Sales Prediction

We applied several different forecasting methods to the Adidas dataset, including deep neural networks. I have listed the results below. It is clear that XGBoost is the relatively optimal solution. The reason for this is simply that there is too little data; a few thousand data points are insufficient to meet the learning requirements of tens of thousands of parameters. Consequently, the model itself does not learn any time-series features; instead, whilst achieving a high accuracy rate, it merely replicates the previous day's sales figures. Please refer to the code for the relevant data.



6. Recommendations

6.1. Segmentation (RFM Feature Construction & K-Means Clustering)

6.1.1. Prioritise Champions and Loyal High-Spenders for retention

These two segments represent the highest current profit and highest predicted CLV. Retention investment should be concentrated here using non-discount tools: loyalty programme access, early product launches, VIP service tiers, and referral incentives.

6.1.2. Reform the discount strategy using RFM scores

Discounts should be deployed by RFM tier, not broadcast uniformly. At-Risk customers (low R, low F) are the appropriate target for reactivation discounts — they are price-sensitive and disengaged. Customers with high R and F scores should never receive discount-led communications. Implement a discount eligibility rule in the CRM: customers scoring F=4 or F=5 are ineligible for price promotions

6.1.3. Redirect product promotion toward Technology

Furniture lines with negative net margins are destroying value. Promotional budgets should be reallocated toward Technology cross-sell campaigns targeting Champions and Loyal High-Spenders. These customers already demonstrate high spend and profit contribution — Technology promotions reinforce, rather than build, this behaviour.

6.2. LDA (Linear Discriminant Analysis) Recommendations

- **At-Risk customers (Cluster 3):** Deploy targeted "We Miss You" reactivation campaigns with time-limited exclusive discounts. Given the 85.7% recall rate, apply a secondary filter of no purchase in the last 180 days before triggering these campaigns to avoid misclassifying Potential Loyalists as At-Risk.
- **Champions (Cluster 0):** Avoid discounts entirely — their 95.2% recall and high frequency profile confirm they do not need price incentives. Focus on loyalty programme access, early product launches, and referral rewards.
- **Loyal High-Spenders (Cluster 2):** Given the 81.8% recall (the lowest of all clusters), these customers require the most careful handling. Cross-sell and upsell campaigns should be personalised, and premium — a misclassified Loyal High-Spender treated as a Champion may disengage permanently.
- **New customers:** Apply the LDA model from the very first transaction. A high-value first purchase with zero discount scores as a potential Champion — begin nurturing immediately rather than waiting for cluster assignment through accumulated RFM history.

6.3. Customer Lifetime Value (CLV)

6.3.1. Reallocate marketing investment toward top-value customers

The model shows strong value concentration (top 20% $\approx$ 48% of future value). Marketing resources should be decisively shifted toward the top-ranked segments, where impact and ROI are significantly higher.

6.3.2. Move from volume-driven to value-driven targeting

Instead of optimizing for reach or conversions, decisions should be guided by predicted future profit. This ensures that acquisition and retention efforts focus on customers who generate long-term value, not just short-term activity.

6.3.3. Operationalize the model for customer selection across channels

The predictive ranking should become the core selection layer for CRM, paid media, and retail media activation, enabling consistent and scalable targeting of high-value audiences.

6.3.4. Embed continuous optimization into the strategy

As customer behavior evolves, the model should be regularly updated and its outputs monitored to ensure sustained targeting accuracy and maintain performance over time.

6.4. Sales Forecasting Recommendations

- For new product launches, build **seasonal procurement buffers of 20–30% above baseline** in Q4 and Q2, consistent with the peaks identified by the Holt-Winters Chairs benchmark model.
- For Adidas-type inventory, the XGBoost feature importance shows **lag_1 and lag_2 account for over 50% of predictive power** — meaning yesterday and the day before are the strongest sales predictors. Set automated alerts when the **7-day rolling average deviates more than 15%** from the prior week, triggering a procurement review.
- Apply the Holt-Winters benchmark approach to all future new product launches — select the closest existing subcategory as a seasonal reference before finalising launch timing.

7. Measurement Plan

The analytical framework's success will be tracked using KPIs aligned to each stage of the model pipeline.

7.1 Customer Segmentation (RFM + K-Means):

Monitor quarterly cluster stability using silhouette scores (target ≥ 0.35). Track average profit per segment over time to detect migration. Flag if Loyal High-Spenders (Cluster 2) profit share drops below 30%, triggering a re-clustering review.

7.2 LDA Model Performance:

Re-run validation every 6 months on new transaction data. Maintain test accuracy $\geq 90\%$. Monitor Loyal High-Spender recall (currently 81.8%) — if it falls below 78%, recalibrate the model. Track misclassification rates by cluster monthly via CRM segment tagging.

7.3 Customer Lifetime Value (CLV):

Measure Capture@20pct quarterly (current baseline: 48.35%). Track actual vs. predicted profit by Priority tier (A/B/C) at 6-month intervals. Monitor Priority A retention rate — target $\geq 85\%$ annually. Re-train the model when Spearman's rank correlation drops below 0.40.

7.4 Category New-Product Sales Prediction:

Compare actual monthly sales against Holt-Winters forecasts. Track MAPE (Mean Absolute Percentage Error) — target $\leq 20\%$ for seasonal peaks. Review benchmark subcategory selection quarterly to ensure relevance for new launches.

7.5 Sales Forecasting (XGBoost):

Track weekly RMSE against baseline (current: 312,977). Monitor 7-day rolling average deviation — trigger procurement review if deviation exceeds 15% from the prior week. Retrain the model every quarter with updated lag features and holiday calendars.

8. Limitations and Assumptions

Data scope and recency:

The Superstore dataset covers 2015–2018, and Adidas covers 2020–2021. Neither dataset reflects current market conditions. All segmentation, CLV, and forecasting outputs are historically calibrated and may not generalise to post-2021 consumer behaviour, particularly given macroeconomic shifts post-pandemic.

Single-retailer bias:

The Superstore data represents one US retail operation. Customer segments and profit patterns may differ significantly across geographies, industries, or multi-channel environments. Generalisation beyond this context should be done cautiously.

RFM feature assumptions:

RFM assumes that recency, frequency, and monetary value are sufficient proxies for future behaviour. It excludes qualitative factors such as customer satisfaction, brand loyalty, competitor activity, or external market events, all of which could materially affect CLV outcomes.

K-Means clustering limitations:

K-Means assumes spherical clusters of equal variance, which may not reflect the true shape of customer distributions. The Elbow Method provides a heuristic, not a definitive, optimal cluster count. Results may shift with different random seeds or feature scaling approaches.

CLV model limitations:

The rolling snapshot design increases training sample size but introduces temporal leakage risk if not strictly managed. The regression shows moderate MAE (141.68), individual-level profit predictions carry uncertainty: the model is validated for ranking & selection, not exact revenue forecasting.

XGBoost lag dependency:

The dominance of lag_1 and lag_2 features (>50% of predictive power) means the XGBoost model is essentially a sophisticated persistence model. It performs well in stable periods but may underperform during demand shocks, promotions, or supply disruptions not present in the training data.

Deep learning data insufficiency:

LSTM and other deep neural networks were tested on the Adidas dataset, but underperformed XGBoost due to insufficient data volume (~9,648 records). This is an inherent constraint of the dataset, not a model design failure. With larger longitudinal data, DL approaches may outperform.

Returns data completeness:

Only 296 returned orders are recorded. If return rates vary systematically by customer segment — for example, if high-spending customers return more frequently — the CLV and profit figures may still be partially overstated.

Assumption of behavioural stability:

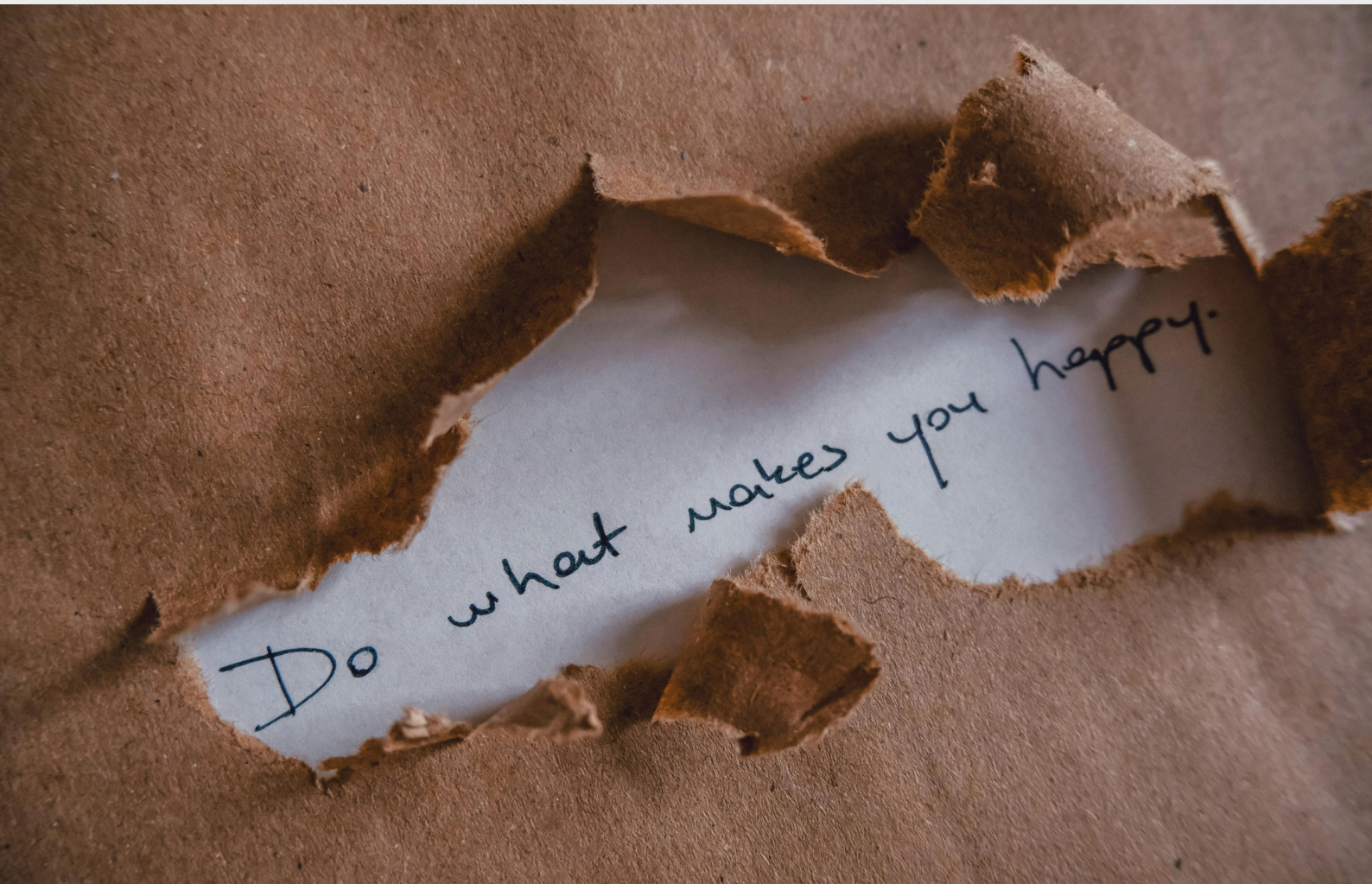
All models assume that past purchasing behaviour is predictive of future behaviour. Significant changes in product mix, pricing strategy, or competitive landscape could invalidate segment assignments and CLV predictions over time.

Contribution Breakdown

Report Contribution Summary

No.	Section
1.	Executive Summary (p.03) — Shushma
2.	Marketing Decision Objectives (p.04) — Shushma
3.	Data Sourcing & Quality (p.05) — Shushma
4.	Methodology & Analysis 4.1 Segmentation (RFM + K-Means) — Ren (p.06) 4.2 Linear Discriminant Analysis (LDA) — Shushma (p.10) 4.3 Customer Lifetime Value (CLV) — Xiaoer (p.13) 4.4 Category New-Product Sales Prediction — Zeyuan (p.15) 4.5 Sales Forecasting — Zeyuan (p.17)
5.	Key Results & Findings 5.1 Segmentation Results — Ren (p.19) 5.2 LDA Results & Findings — Shushma (p.22) 5.3 CLV Results — Xiaoer (p.25) 5.4 New-Product Sales Prediction Results — Zeyuan 5.5 Sales Forecasting Results — Zeyuan (p.29)
6.	Recommendations (p.30) Joint Contribution — Ren, Shushma, Xiaoer, Zeyuan
7.	Measurement Plan (p.35) — Shushma
8.	Limitations & Assumptions (p.35) — Shushma

Thank You!



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